- **Continuous-Time Fourier Series (CTFS) Pair**

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \tag{3.38}$$

$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_T x(t)\, e^{-jk(2\pi/T)t}\, dt \tag{3.39}$$

Eq.(3.38) : synthesis equation
Eq.(3.39) : analysis equation
$\{a_k\}$: Fourier series coefficients or spectral coefficients of $x(t)$
a_0 : DC component of $x(t)$ ($a_0 = \frac{1}{T}\int_0^T x(t)\, dt$) [average value]

(Example 3.4)

$$x(t) = 1 + \sin\omega_0 t + 2\cos\omega_0 t + \cos(2\omega_0 t + \pi/4)$$

$$= 1 + (1 + \frac{1}{2j})e^{j\omega_0 t} + (1 - \frac{1}{2j})e^{-j\omega_0 t} + (\frac{1}{2}e^{j(\pi/4)})e^{j2\omega_0 t} + (\frac{1}{2}e^{-j(\pi/4)})e^{-j2\omega_0 t}$$

$$\equiv \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Thus, the Fourier series coefficients

$$a_0 = 1$$

$$a_1 = 1 + \frac{1}{2j} = 1 - \frac{1}{2}j$$

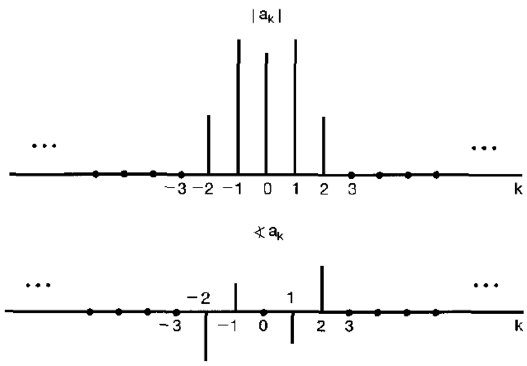
$$a_{-1} = 1 - \frac{1}{2j} = 1 + \frac{1}{2}j$$

$$a_2 = \frac{1}{2}e^{j(\pi/4)} = \frac{\sqrt{2}}{4}(1+j)$$

$$a_{-2} = \frac{1}{2}e^{-j(\pi/4)} = \frac{\sqrt{2}}{4}(1-j)$$

$$a_k = 0, \quad |k| > 2$$

a bar graph of the magnitude and phase of a_k [the spectrum of $x(t)$]



※ other representation method for a sinewave using 2 bars, that is, the representation method in the frequency domain.

(Example 3.5) a periodic square wave

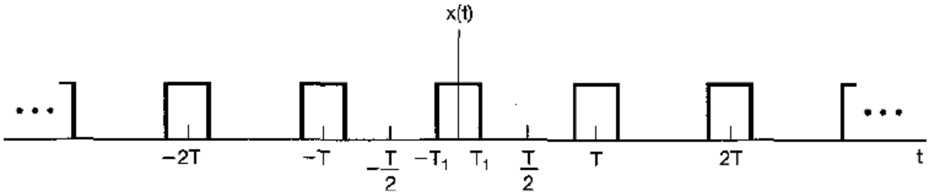


Figure 3.6 Periodic square wave.

For $k=0$,

$$a_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)\, dt = \frac{1}{T}\int_{-T_1}^{T_1} 1\ dt = \frac{2T_1}{T}$$

For $k \neq 0$,

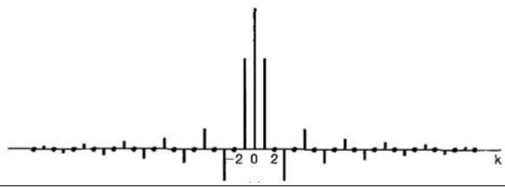
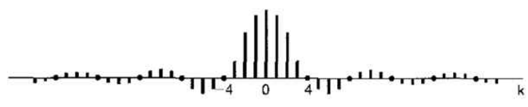
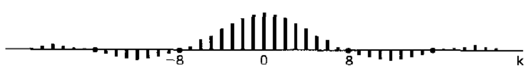
$$a_k = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)\, e^{-jk\omega_0 t} dt = \frac{1}{T}\int_{-T_1}^{T_1} 1\ e^{-jk\omega_0 t}\ dt$$

$$= -\frac{1}{jk\omega_0 T}\, e^{-jk\omega_0 t}\Big|_{-T_1}^{T_1}$$

$$= \frac{2}{k\omega_0 T}\left[\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j}\right]$$

$$= \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \frac{\sin(k\omega_0 T_1)}{k\pi}, \quad k \neq 0 \qquad (3.44)$$

Plots of the Fourier series coefficients $T a_k$

$T=4T_1$ (period)=2(pulse width)	
$T=8T_1$ (period)=4(pulse width)	
$T=16\,T_1$ (period)=8(pulse width)	

the envelope : $(2\sin\omega T_1)/\omega$

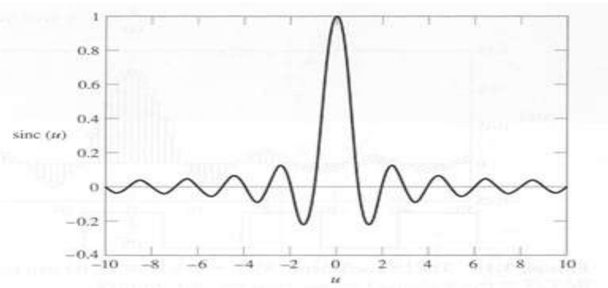
the spacing between samples : $2\pi/T$

the first zero-crossing : 2nd / 4th / 8th

※ Since $a_0 = \lim_{k \to 0} \dfrac{2\sin(k\omega_0 T_1)}{k\omega_0 T} = \dfrac{2T_1}{T}$,

a_k can be represented $a_k = \dfrac{2\sin(k\omega_0 T_1)}{k\omega_0 T}$

※ sinc function : $\mathrm{sinc}(u) = \dfrac{\sin(\pi u)}{\pi u}$



main lobe : between the first zero crossings
side lobes : outsides of the first zero crossing

※ Eq.(3.44) can be represented using sinc function

$$a_k = \frac{2\sin(2k\pi \frac{T_1}{T})}{2k\pi} = \frac{2T_1}{T}\frac{\sin(2k\pi\frac{T_1}{T})}{2\pi k\frac{T_1}{T}} = \frac{2T_1}{T}\mathrm{sinc}(2k\frac{T_1}{T})$$

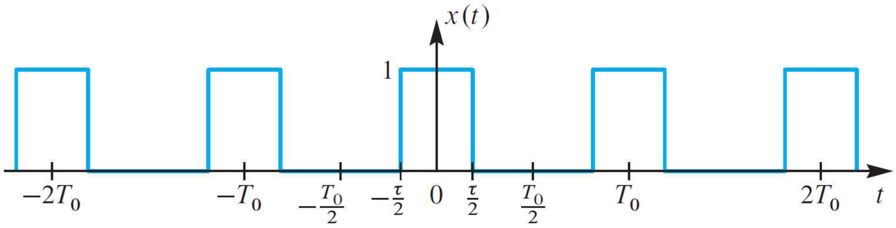
$$= \frac{\tau}{T}\frac{\sin(k\pi\frac{\tau}{T})}{k\pi\frac{\tau}{T}} = \frac{\tau}{T}\mathrm{sinc}(k\frac{\tau}{T})$$

(let $\tau = 2T_1$)

τ : pulse width

(※ Duty Cycle $\dfrac{\tau}{T} \times 100\ [\%]$)

[spectrum using the pulse width (τ)]



Fourier coefficients of the pulse train (period $T_0 = T$)

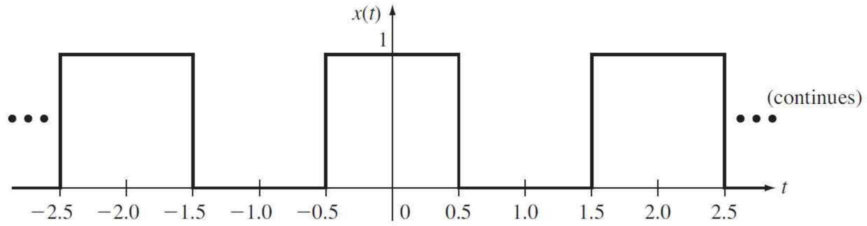
$$a_k = \frac{\tau}{T}\frac{\sin(k\pi\frac{\tau}{T})}{k\pi\frac{\tau}{T}} = \frac{\tau}{T}\,\mathrm{sinc}(k\frac{\tau}{T})$$

Duty cycle ($\frac{\tau}{T}$)	a_k
$\frac{1}{2}$	$\frac{1}{2}\frac{\sin(k\pi\frac{1}{2})}{k\pi\frac{1}{2}} = \frac{\sin(\frac{k\pi}{2})}{k\pi}$
$\frac{1}{4}$	$\frac{1}{4}\frac{\sin(k\pi\frac{1}{4})}{k\pi\frac{1}{4}} = \frac{\sin(\frac{k\pi}{4})}{k\pi}$
$\frac{1}{8}$	$\frac{1}{8}\frac{\sin(k\pi\frac{1}{8})}{k\pi\frac{1}{8}} = \frac{\sin(\frac{k\pi}{8})}{k\pi}$
$\frac{1}{16}$	$\frac{1}{16}\frac{\sin(k\pi\frac{1}{16})}{k\pi\frac{1}{16}} = \frac{\sin(\frac{k\pi}{16})}{k\pi}$

spectrum interval : fundamental frequency $f_0 = \frac{1}{T}$ [Hz]

first zero crossing frequency : $\frac{1}{\tau}$ [Hz] (inverse of the pulth width)

(Example) Rectangular Pulse Train [* Kamen book *]



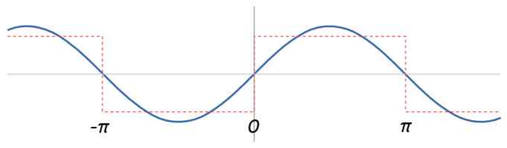
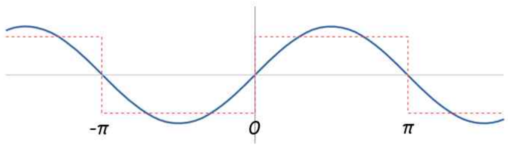
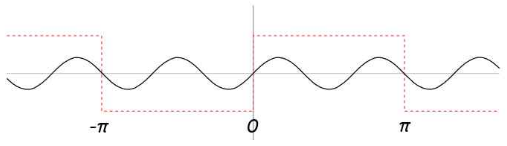
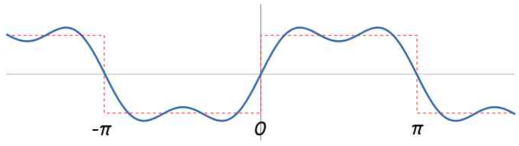
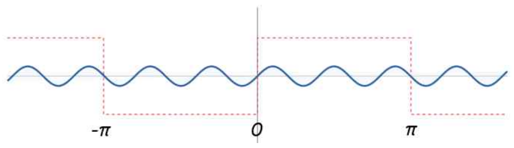
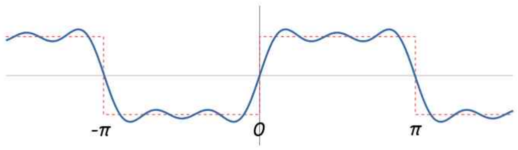
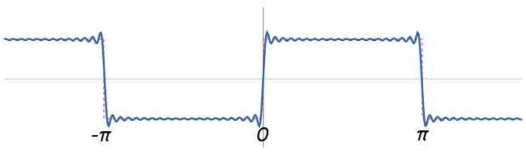
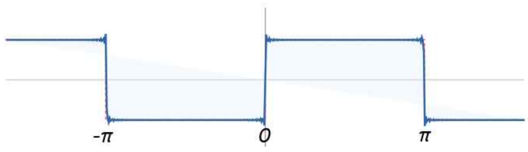
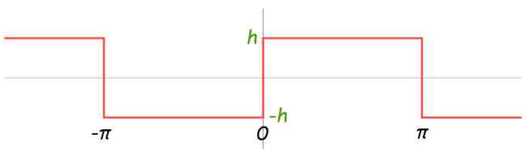
Sinusoidal form

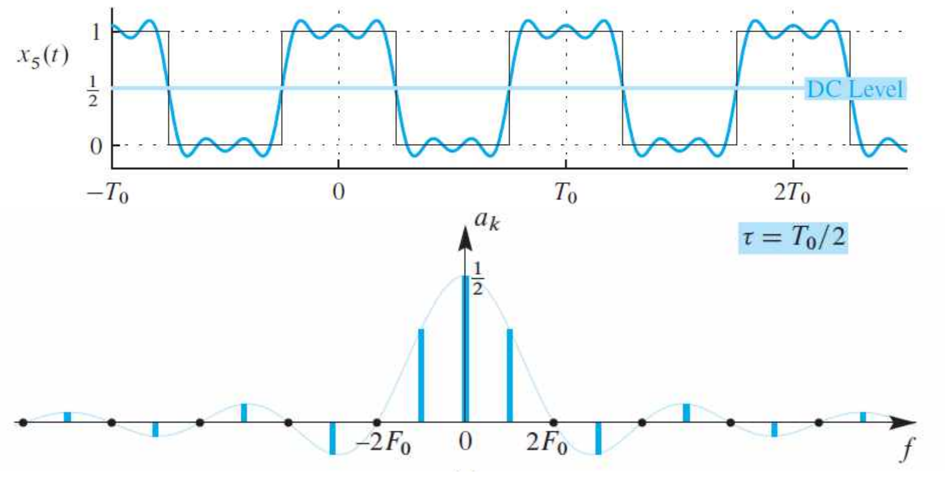
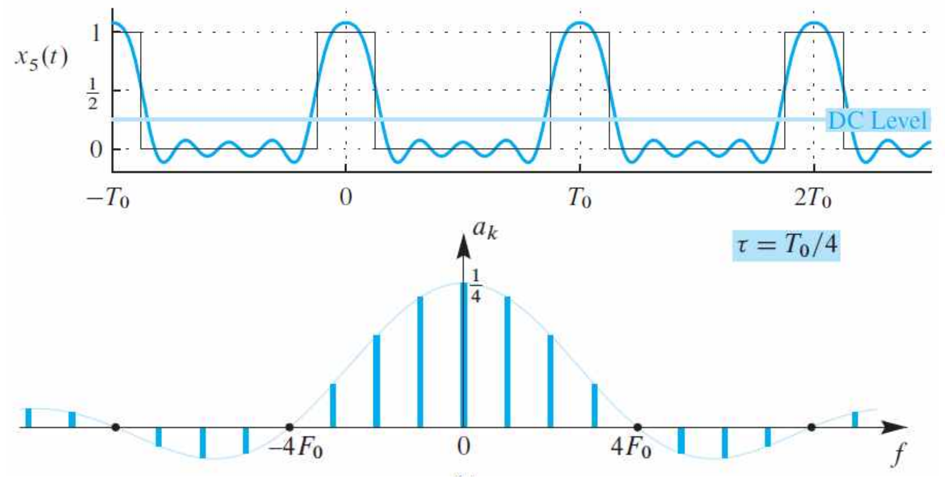
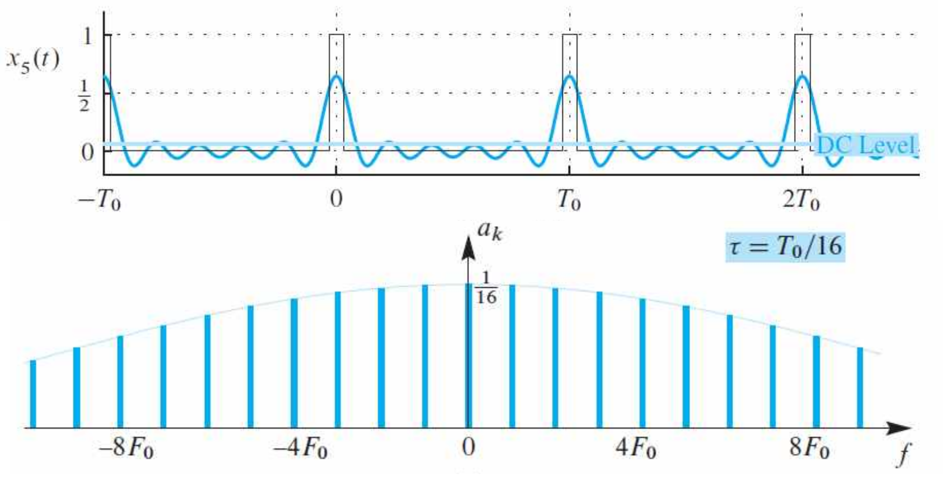
$$x(t) = \frac{1}{2} + \frac{2}{\pi}\cos(\pi t) - \frac{2}{3\pi}\cos(3\pi t) + \frac{2}{5\pi}\cos(5\pi t) - \frac{2}{7\pi}\cos(7\pi t) + \cdots$$

Complex exponential form

$$a_k = \frac{\sin\left(\frac{k\pi}{2}\right)}{k\pi}$$

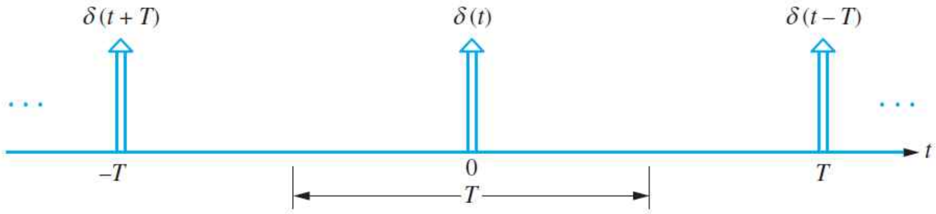
※ Can we make a square wave by adding sinusoidal waves? [YES !!]

Start with **sin(x)**:  	Start with **sin(x)**: 
Then take **sin(3x)/3**:  	And add it to make **sin(x)+sin(3x)/3**: 
Now take **sin(5x)/5**:  	Add it also, to make **sin(x)+sin(3x)/3+sin(5x)/5**: 
. . . .	. . . .
Using 20 sine waves we get **sin(x)+sin(3x)/3+sin(5x)/5 + ... + sin(39x)/39**	
Using 100 sine waves we get **sin(x)+sin(3x)/3+sin(5x)/5 + ... + sin(199x)/199**	
a square wave = sin(x) + sin(3x)/3 + sin(5x)/5 + ... (infinitely)	

<table>
<tr><th>duty cycle</th><th>spectrum (a_k)</th></tr>
<tr><td>$\frac{1}{2}$</td><td>(a) Pulse Width: $\tau = T_0/2$
</td></tr>
<tr><td>$\frac{1}{4}$</td><td>(b) Pulse Width: $\tau = T_0/4$
</td></tr>
<tr><td>$\frac{1}{16}$</td><td>(c) Pulse Width: $\tau = T_0/16$
</td></tr>
</table>

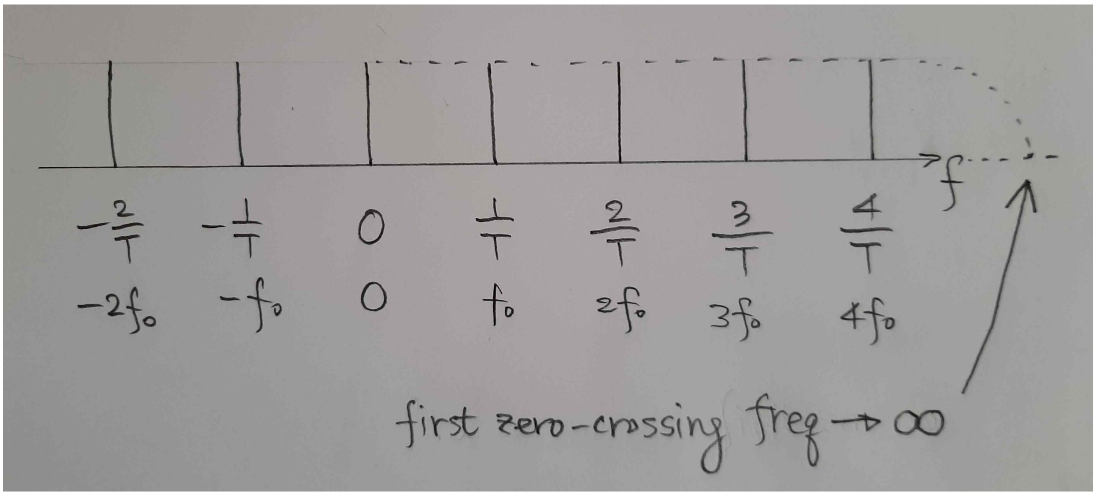
(Impulse Train) pulse with (τ) $\longrightarrow$ 0

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

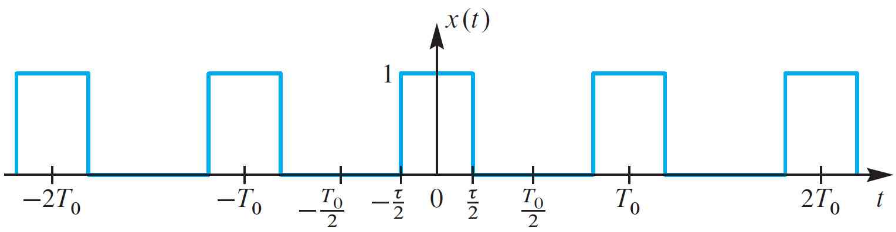


- Spectrum of a impulse train

first zero-crossing frequency ($\frac{1}{\tau}$) $\longrightarrow$ ∞



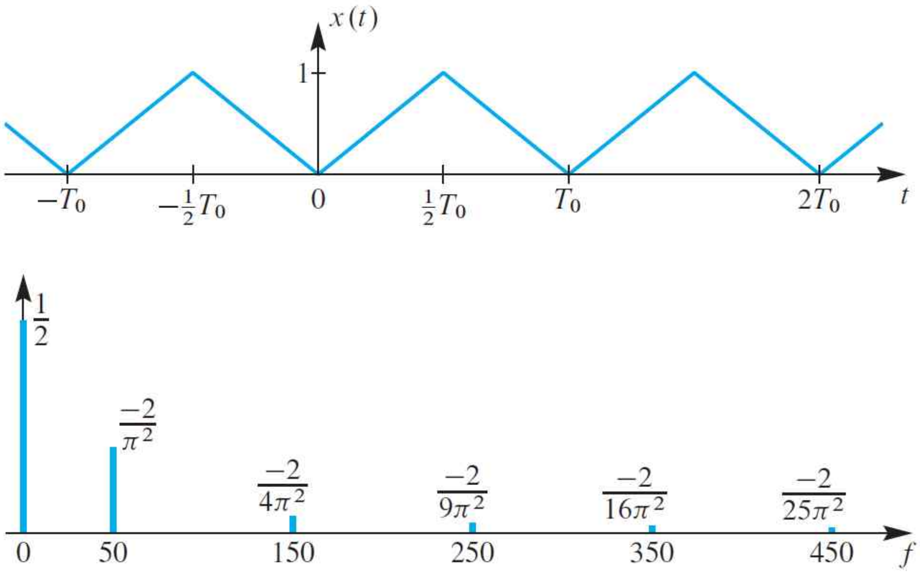
(Think about the spectrum)



(i) pulse width (τ) fixed, period (T_0) $\longrightarrow$ ∞ [nonperiodic]

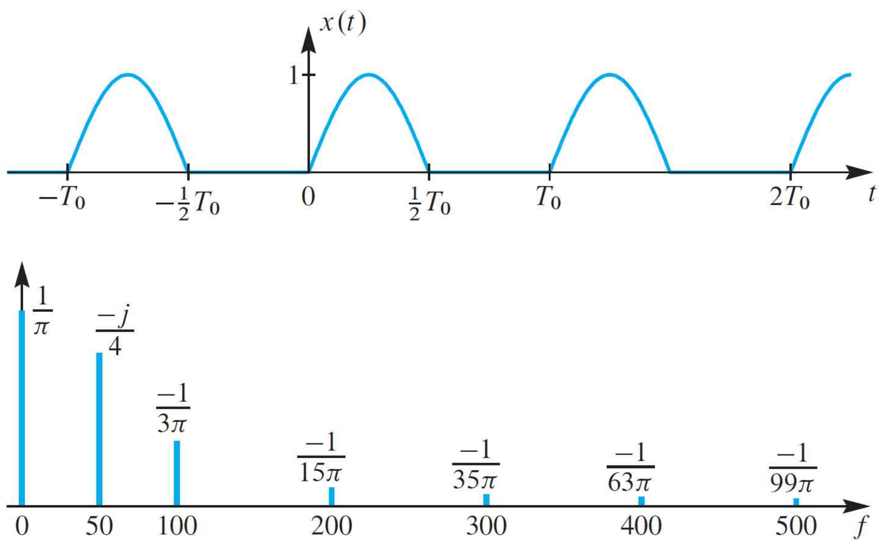
(ii) pulse with (τ) $\longrightarrow$ 0 , period (T_0) $\longrightarrow$ ∞ [impulse]

(Spectrum of a Triangular Wave) [$T_0 = 0.02$ sec]



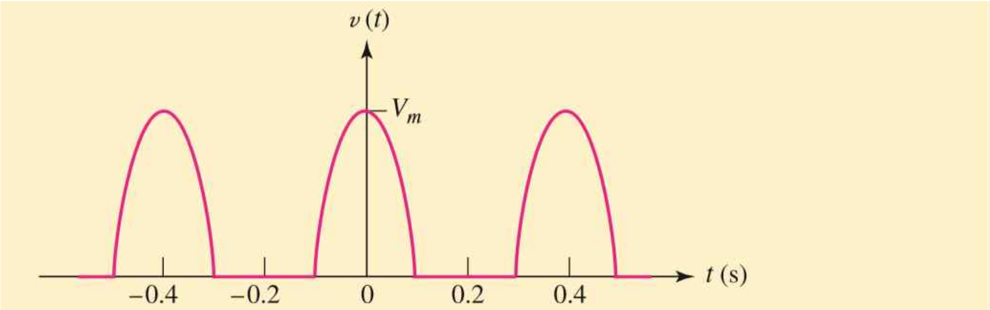
(※ even function, so spectra are real numbers)

(Spectrum of a Half-wave Rectified Sinewave) [$T_0 = 0.02$ sec]



(※ not even function, so spectra are complex numbers)

※ MIT Open Courseware
https://ocw.mit.edu/courses/6-003-signals-and-systems-fall-2011/resources/mit6_003f11_lec14/



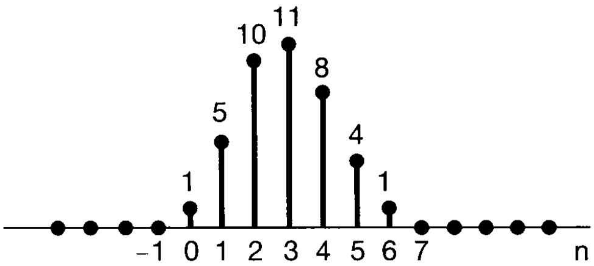
$$v(t) = \frac{V_m}{\pi} + \frac{V_m}{2}\cos 5\pi t + \frac{2V_m}{3\pi}\cos 10\pi t - \frac{2V_m}{15\pi}\cos 20\pi t + \frac{2V_m}{35\pi}\cos 30\pi t - \cdots \qquad [16]$$

● **Sifting Property**

- (Chapter 2) Sifting Property in TIME DOMAIN

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\ \delta[n-k]$$

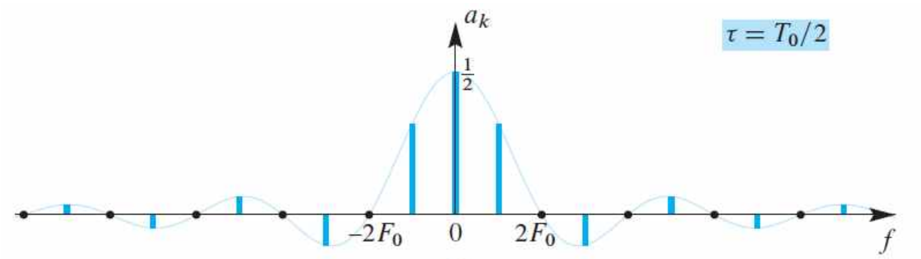
(Example)



$$\begin{aligned} x[n] &= x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + x[3]\delta[n-3] \\ &\quad + x[4]\delta[n-4] + x[5]\delta[n-5] + x[6]\delta[n-6] \\ &= 1\,\delta[n] + 5\,\delta[n-1] + 10\,\delta[n-2] + 11\,\delta[n-3] \\ &\quad + 8\,\delta[n-4] + 4\,\delta[n-5] + 1\,\delta[n-6] \end{aligned}$$

- (Chapter 3) Sifting Property in FREQUENCY DOMAIN

(Example) **Fourier Series**



$$\begin{aligned} X[k] &= \cdots + a_{-3}\,\delta[k-(-3)] + a_{-2}\,\delta[k-(-2)] + a_{-1}\,\delta[k-(-1)] \\ &\quad + a_0\,\delta[k] + a_1\,\delta[k-1] + a_2\,\delta[k-2] + a_3\,\delta[k-3] + \cdots \\ X(\omega) &= \cdots + a_{-3}\,\delta[\omega + 3\omega_0] + a_{-2}\,\delta[\omega + 2\omega_0] + a_{-1}\,\delta[\omega + \omega_0] \\ &\quad + a_0\,\delta[\omega] + a_1\,\delta[\omega - \omega_0] + a_2\,\delta[\omega - 2\omega_0] + a_3\,\delta[\omega - 3\omega_0] + \cdots \end{aligned}$$

※ (Chapter 2) A signal is a linear combination of delta functions in TIME DOMAIN.

(Chapter 3) A signal is a linear combination of delta functions in FREQUENCY DOMAIN. [Fourier Series]

(Chapter 2 & 3) A signal is a linear combination of delta functions in TIME and FREQUENCY DOMAINs. [sifting property]

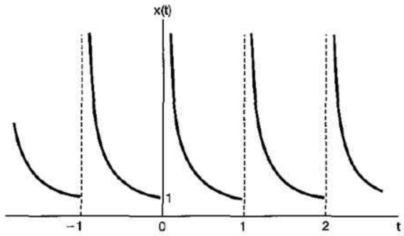
§3.4 Convergence of the Fourier series

- The Dirichlet conditions for the convergence of the FS

[Condition 1] Over any period, $x(t)$ must be absolutely integrable.

$$\int_T |x(t)|\, dt < \infty$$

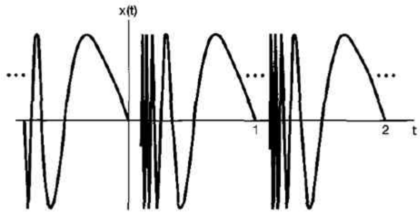
(ex) violating condition 1



[Condition 2] In any finite interval of time, $x(t)$ is of bounded variation.

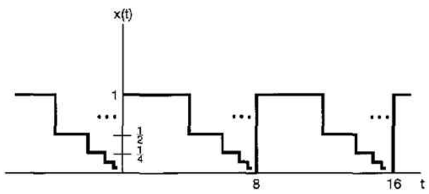
(ex) violating condition 2

$$x(t) = \sin\left(\frac{2\pi}{t}\right), \quad 0 < t \leqq 1$$



[Condition 3] In any finite interval of time, there are a finite number of discontinuities.

(ex) violating condition 3



※ Gibbs Phenomenon

- 1898 : Albert Michelson constructed a harmonic analyzer.
- Fourier series

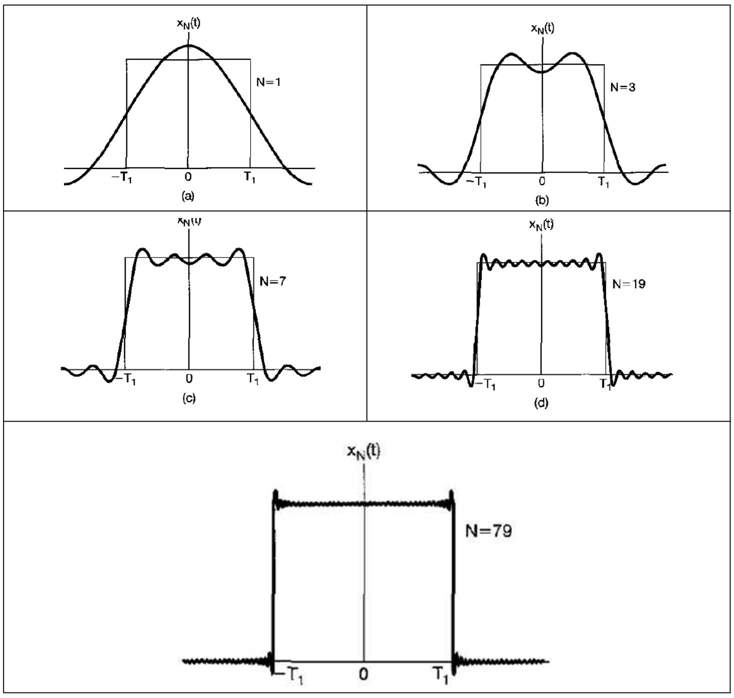
(infinite number of complex exponentials)

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$$

Michelson's harmonic analyzer in the real world

(finite number of complex exponentials)

$$x_N(t) = \sum_{k=-N}^{+N} a_k e^{jk\omega_0 t}$$



※ As N increases, the ripples in the partial sums become compressed toward the discontinuity, but for any finite value of N, no matter how large N becomes, the peak amplitude of the ripples remains constant, 1.09, (i.e., an overshoot of 9% of the height of the discontinuity.

§3.5 Properties of continuous-time Fourier Series (CTFS)

- the notation of CTFS

$$x(t) \leftrightarrow a_k$$

3.5.1 Linearity

$$x(t) \leftrightarrow a_k$$
$$y(t) \leftrightarrow b_k$$
$$z(t) \leftrightarrow c_k$$

$$z(t) = Ax(t) + By(t) \quad \leftrightarrow \quad c_k = Aa_k + Bb_k$$

3.5.2 Time Shifting

$$x(t-t_0) \quad \leftrightarrow \quad e^{-jk\omega_0 t_0}\, a_k = e^{-jk(2\pi/T)\omega_0}\, a_k$$

[time delay in time domain ≡ phase shift in frequency domain]

$$|\, e^{-jk\omega_0 t_0}\, a_k \,| = |\, a_k \,|$$

3.5.3 Time Reversal

$$x(-t) \quad \leftrightarrow \quad a_{-k}$$

3.5.4 Time Scaling

$$x(\alpha t) = \sum_{k=-\infty}^{+\infty} a_k\, e^{jk(\alpha\omega_0)t}$$

※ The Fourier coefficients are not changed, BUT the Fourier series representation is changed because of the change of the fundamental frequency ($\omega_0 \rightarrow \alpha\omega_0$).

3.5.5 Multiplication (Modulation)

$$x(t)\, y(t) \quad \leftrightarrow \quad \sum_{l=-\infty}^{+\infty} a_l\, b_{k-l}$$

[multiplication of two time functions ≡ convolution in frequency domain]
[convolution in time domain ≡ multiplication of two frequency functions]

※ duality

3.5.6 Complex Conjugate Symmetry

$x(t)$: real → $a_{-k} = a_k^*$

$x(t)$: real and even → $a_{-k} = a_k$

3.5.7 Parseval's Relation

$$\frac{1}{T}\int_T |x(t)|^2\,dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

$$\left\{\begin{matrix}\text{average power}\\ \text{in time domain}\end{matrix}\right\} = \left\{\begin{matrix}\text{average power}\\ \text{in frequency domain}\end{matrix}\right\}$$

3.5.8 Summary

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$x(t)$, $y(t)$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k, b_k
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} x(t) = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a^*_{-k}
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\frac{2\pi}{T}a_k$
Integration		$\int_{-\infty}^{t} x(t)\,dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a^*_{-k} \\ \mathcal{Re}\{a_k\} = \mathcal{Re}\{a_{-k}\} \\ \mathcal{Im}\{a_k\} = -\mathcal{Im}\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real]	$\mathcal{Re}\{a_k\}$
		$x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$j\mathcal{Im}\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

※ Even if the amplitude and the frequency of the sine waves in a signal are same, if the phases are different, the resultant signals are absolutely different signals. In other words, if the phases are different, we can make several waves.

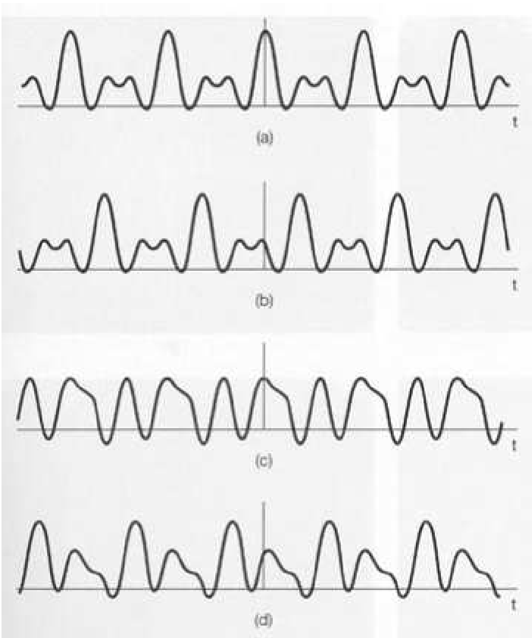


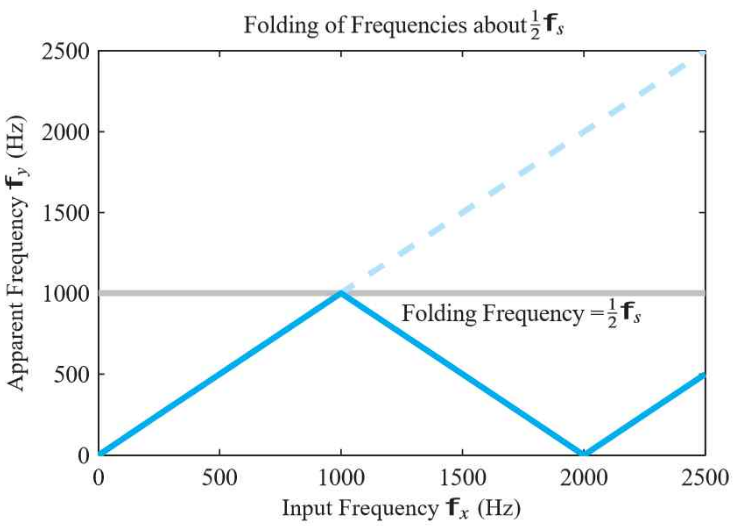
Figure 6.1 The signal $x(t)$ given in eq. (6.3) for several different choices of the phase angles ϕ_1, ϕ_2, and ϕ_3: (a) $\phi_1 = \phi_2 = \phi_3 = 0$; (b) $\phi_1 = 4$ rad, $\phi_2 = 8$ rad, $\phi_3 = 12$ rad; (c) $\phi_1 = 6$ rad, $\phi_2 = -2.7$ rad, $\phi_3 = 0.93$ rad; (d) $\phi_1 = 1.2$ rad, $\phi_2 = 4.1$ rad, $\phi_3 = -7.02$ rad.

$$x(t) = 1 + \frac{1}{2}\cos(2\pi t + \phi_1) + \cos(4\pi t + \phi_2) + \frac{2}{3}\cos(6\pi t + \phi_3)$$

§3.6 Fourier series representation of Discrete-Time Periodic Signal (DTFS)

§3.6.1 Linear Combinations of harmonically related complex exponentials

Fourier series	CTFS	DTFS
periodic function	$x(t)=x(t+T)$	$x[n]=x[n+N]$
frequency	$\omega_0=2\pi/T$	$\omega_0=2\pi/N$
basic signals	$e^{j\omega_0 t},\ e^{j2\omega_0 t},\ e^{j3\omega_0 t},\ \cdots$	$e^{j\omega_0 n},\ e^{j2\omega_0 n},\ e^{j3\omega_0 n},\ \cdots$
linear combination of basic signals (harmonically related complex exponentials)	$x(t)=\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$ $=\sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$	$x[n]=\sum_{k=<N>} a_k e^{jk\omega_0 n}$ $=\sum_{k=<N>} a_k e^{jk(2\pi/N)n}$
remark	$e^{j(\omega_0+2\pi)t} \neq e^{j\omega_0 t}$	$e^{j(\omega_0+2\pi)n} = e^{j\omega_0 n}$
summation interval	$-\infty<k<+\infty$ [infinite number of complex exponentials to represent $x(t)$]	$<N>$ [finite number(N) of complex exponentials to represent $x[n]$]



3.6.2 Determination of the coefficients of DTFS

- Assuming that a given periodic signal $x[n]$ with the fundamental frequency ω_0 can be represented with the series of Eq.(3.88), we need a procedure for determining the coefficients a_k [Fourier coefficients]

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} = \sum_{k=<N>} a_k e^{j2\pi kn/N} \quad (3.88)$$

$$\begin{pmatrix} \text{determination of the} \\ \text{Fourier coefficients, } a_k \end{pmatrix} \equiv \begin{pmatrix} \text{finding a solution to} \\ \text{a set of linear equations} \end{pmatrix}$$

- Determination of the Fourier coefficients, a_k

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} = \sum_{k=<N>} a_k e^{j2\pi kn/N}$$

$$\begin{aligned}
x[0] &= \sum_{k=<N>} a_k \\
x[1] &= \sum_{k=<N>} a_k e^{j2\pi k(1)/N} \\
x[2] &= \sum_{k=<N>} a_k e^{j2\pi k(2)/N} \qquad (3.89) \\
&\cdots\cdots\cdots \\
x[N-1] &= \sum_{k=<N>} a_k e^{j2\pi k(N-1)/N} \\
&\left(= \sum_{k=<N>} a_k e^{j2\pi k} e^{j2\pi k(-1)/N} = x[-1]\right)
\end{aligned}$$

$$\begin{aligned}
x[N] &= \sum_{k=<N>} a_k e^{j2\pi k(N)/N} = \sum_{k=<N>} a_k e^{j2\pi k} = \sum_{k=<N>} a_k = x[0] \\
x[N+1] &= \sum_{k=<N>} a_k e^{j2\pi k(N+1)/N} = \sum_{k=<N>} a_k e^{j2\pi k} e^{j2\pi k(1)/N} = x[1] \\
x[N+2] &= \cdots\cdots = x[2] \\
&\cdots\cdots\cdots \\
x[2N-1] &= \cdots\cdots = x[N-1] = x[-1]
\end{aligned}$$

$$\begin{aligned}
x[2N] &= \cdots\cdots = x[0] \\
&\cdots\cdots\cdots
\end{aligned}$$

Repeat and Repeat ………

- The number of independent equations is N.
  Thus, Eq.(3.89) represents a set of N linear equations for the N unknown coefficients a_k as k ranges over a set of N successive integers.

$$\begin{matrix} N \text{ given values} \\ \text{of } x[n] \end{matrix} \xrightarrow[\text{Eq(3.89)}]{\substack{\text{solve } N \text{ linearly} \\ \text{independent equations}}} \begin{matrix} \text{obtain } N \text{ unknown} \\ \text{coefficients } a_k \end{matrix}$$

Rewrite Eq.(3.88)

$$x[n] = \sum_{k=<N>} a_k e^{j2\pi kn/N} = \sum_{k=0}^{N-1} a_k e^{j2\pi kn/N} \qquad (3.88)$$

Multiplying both sides by $e^{-jr(2\pi/N)n}$ and summing over N terms,

$$\sum_{n=0}^{N-1} x[n] e^{-jr(2\pi/N)n} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} a_k \, e^{j(k-r)(2\pi/N)n} \qquad (3.91)$$

Interchanging the order of summation on the right-hand side,

$$\sum_{n=0}^{N-1} x[n] e^{-jr(2\pi/N)n} = \sum_{k=0}^{N-1} a_k \sum_{n=0}^{N-1} e^{j(k-r)(2\pi/N)n} \qquad (3.92)$$

Since

$$\sum_{n=0}^{N-1} e^{jk(2\pi/N)n} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \cdots \\ 0, & \text{otherwise} \end{cases} \qquad (3.90)$$

Therefore, the innermost sum on the right-hand side of Eq.(3.92)

$$\sum_{n=0}^{N-1} e^{j(k-r)(2\pi/N)n} = \begin{cases} N, & k = r \\ 0, & k \neq r \end{cases}$$

From Eq.(3.92), Fourier coefficients are obtained

$$a_r = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jr(2\pi/N)n} \qquad (3.93)$$

- the Discrete-time Fourier series (DTFS) pair

$$\boxed{\begin{aligned} x[n] &= \sum_{k=<N>} a_k e^{jk\omega_0 n} = \sum_{k=<N>} a_k e^{jk(2\pi/N)n} && (3.94) \\ a_k &= \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk(2\pi/N)n} && (3.95) \end{aligned}}$$

※ the continuous-time Fourier series (CTFS) pair

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t} \qquad (3.38)$$

$$a_k = \frac{1}{T} \int_T x(t) \, e^{-jk\omega_0 t} \, dt = \frac{1}{T} \int_T x(t) \, e^{-jk(2\pi/T)t} \, dt \qquad (3.39)$$

Eq.(3.94), (3.38) : the synthesis equations

Eq.(3.95), (3.39) : the analysis equations

a_k : the spectral coefficients of $x[n]$ or $x(t)$

※ CTFS vs. DTFS

CTFS	DTFS
$x(t)=\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$x[n]=\sum_{k=<N>} a_k e^{jk\omega_0 n}$
INFINITE series	FINITE series

- Repetition of the DTFS coefficients, a_k

Eq.(3.95): $a_k=\frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\omega_0 n}=\frac{1}{N}\sum_{n=<N>} x[n]e^{-jk(2\pi/N)n}$

$$a_0=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j0(2\pi/N)n}=\frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

$$a_1=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j1(2\pi/N)n}$$

$$a_2=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j2(2\pi/N)n}$$

………

$$a_{N-2}=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N-2)(2\pi/N)n}$$
$$=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N)(2\pi/N)n}e^{-j(-2)(2\pi/N)n}=a_{-2}$$

$$a_{N-1}=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N-1)(2\pi/N)n}$$
$$=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N)(2\pi/N)n}e^{-j(-1)(2\pi/N)n}=a_{-1}$$

$$a_N=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N)(2\pi/N)n}=\frac{1}{N}\sum_{n=0}^{N-1} x[n]=a_0$$

$$a_{N+1}=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N+1)(2\pi/N)n}$$
$$=\frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j(N)(2\pi/N)n}e^{-j(1)(2\pi/N)n}=a_1$$

………

$$a_{2N-1} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,e^{-j(2N-1)(2\pi/N)n}$$
$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,e^{-j(2N)(2\pi/N)n}\,e^{-j(-1)(2\pi/N)n} = a_{-1}$$
$$a_{2N} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,e^{-j(2N)(2\pi/N)n} = a_0$$
$$a_{2N+1} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,e^{-j(2N+1)(2\pi/N)n}$$
$$= \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,e^{-j(2N)(2\pi/N)n}\,e^{-j(1)(2\pi/N)n} = a_1$$

………

Repeat and Repeat ………

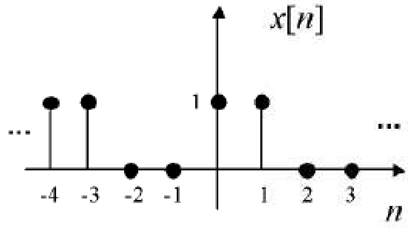
Thus, if we consider more than N sequential values of k, the values a_k repeat periodically with period N.

$$a_k = a_{\pm N+k} = a_{\pm 2N+k} = a_{\pm 3N+k} = \cdots\cdots$$
$$a_{-k} = a_{\pm N-k} = a_{\pm 2N-k} = a_{\pm 3N-k} = \cdots\cdots$$

- Since there are only N DISTINCT complex exponentials that are periodic with period N, the DTFS representation is a FINITE series with N terms. (※ differences with the CTFS)

$$\begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix} = \frac{1}{N}\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{-j\frac{2\pi}{N}} & e^{-j\frac{2\pi}{N}2} & \cdots & e^{-j\frac{2\pi}{N}(N-1)} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & e^{-j(N-1)\frac{2\pi}{N}} & e^{-j(N-1)\frac{2\pi}{N}2} & \cdots & e^{-j(N-1)\frac{2\pi}{N}(N-1)} \end{bmatrix}\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}.$$

(Example) N=4, find the Fourier coefficients, a_0, a_1, a_2 and a_3 .



$$a_0 = \frac{1}{4}\sum_{n=0}^{3} x[n] = \frac{1}{2}$$

$$a_1 = \frac{1}{4}\sum_{n=0}^{3} x[n]e^{-j\frac{2\pi}{4}n} = \frac{1}{4}\left(1+e^{-j\frac{\pi}{2}}\right) = \frac{1}{4}(1-j) = \frac{1}{2\sqrt{2}}e^{-j\frac{\pi}{4}}$$

$$a_2 = \frac{1}{4}\sum_{n=0}^{3} x[n]e^{-j2\frac{2\pi}{4}n} = \frac{1}{4}\left(1+e^{-j\pi}\right) = \frac{1}{4}(1-1) = 0$$

$$a_3 = \frac{1}{4}\sum_{n=0}^{3} x[n]e^{-j3\frac{2\pi}{4}n} = \frac{1}{4}\left(1+e^{-j\frac{3\pi}{2}}\right) = \frac{1}{4}(1+j) = \frac{1}{2\sqrt{2}}e^{+j\frac{\pi}{4}}$$

Let us see if we can recover $x[1]$ using the synthesis equation:

$$x[1] = \sum_{k=0}^{3} a_k e^{jk\frac{2\pi}{4}} = \frac{1}{2} + \frac{1}{2\sqrt{2}}e^{-j\frac{\pi}{4}}e^{j\frac{\pi}{2}} + 0 + \frac{1}{2\sqrt{2}}e^{j\frac{\pi}{4}}e^{-j\frac{\pi}{2}} = \frac{1}{2} + \frac{1}{\sqrt{2}}\mathrm{Re}\{e^{j\frac{\pi}{4}}\} = 1$$

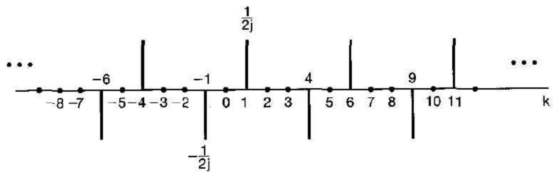
(Example 3.10)

$$x[n] = \sin\omega_o n = \sin(2\pi/N)n = \frac{1}{2j}e^{j(2\pi/N)n} - \frac{1}{2j}e^{-j(2\pi/N)n}$$

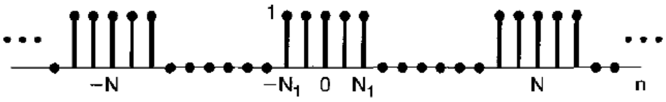
Therefore,

$$a_1 = \frac{1}{2j} \quad , \qquad a_{-1} = -\frac{1}{2j}$$

If $N = 5$, then



(Example 3.12) Discrete-time periodic square wave (Fig.3.16)



Using Eq.(3.95)

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk(2\pi/N)n}$$

$$= \frac{1}{N} \sum_{n=-N_1}^{N_1} 1 \; e^{-jk(2\pi/N)n}$$

Letting $m = n + N_1$

$$a_k = \frac{1}{N} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)(m-N_1)} \qquad (3.103)$$

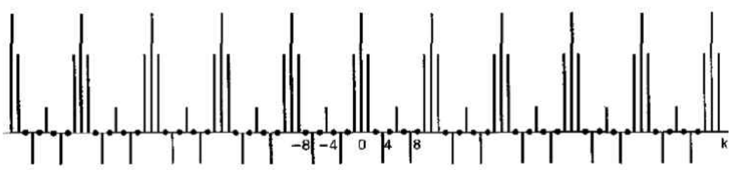
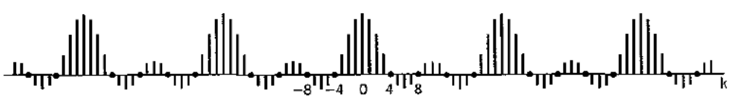
$$= \frac{1}{N} e^{jk(2\pi/N)N_1} \sum_{m=0}^{2N_1} e^{-jk(2\pi/N)m}$$

$$= \frac{1}{N} e^{jk(2\pi/N)N_1} \left(\frac{1-e^{-jk2\pi(2N_1+1)/N}}{1-e^{-jk(2\pi/N)}} \right)$$

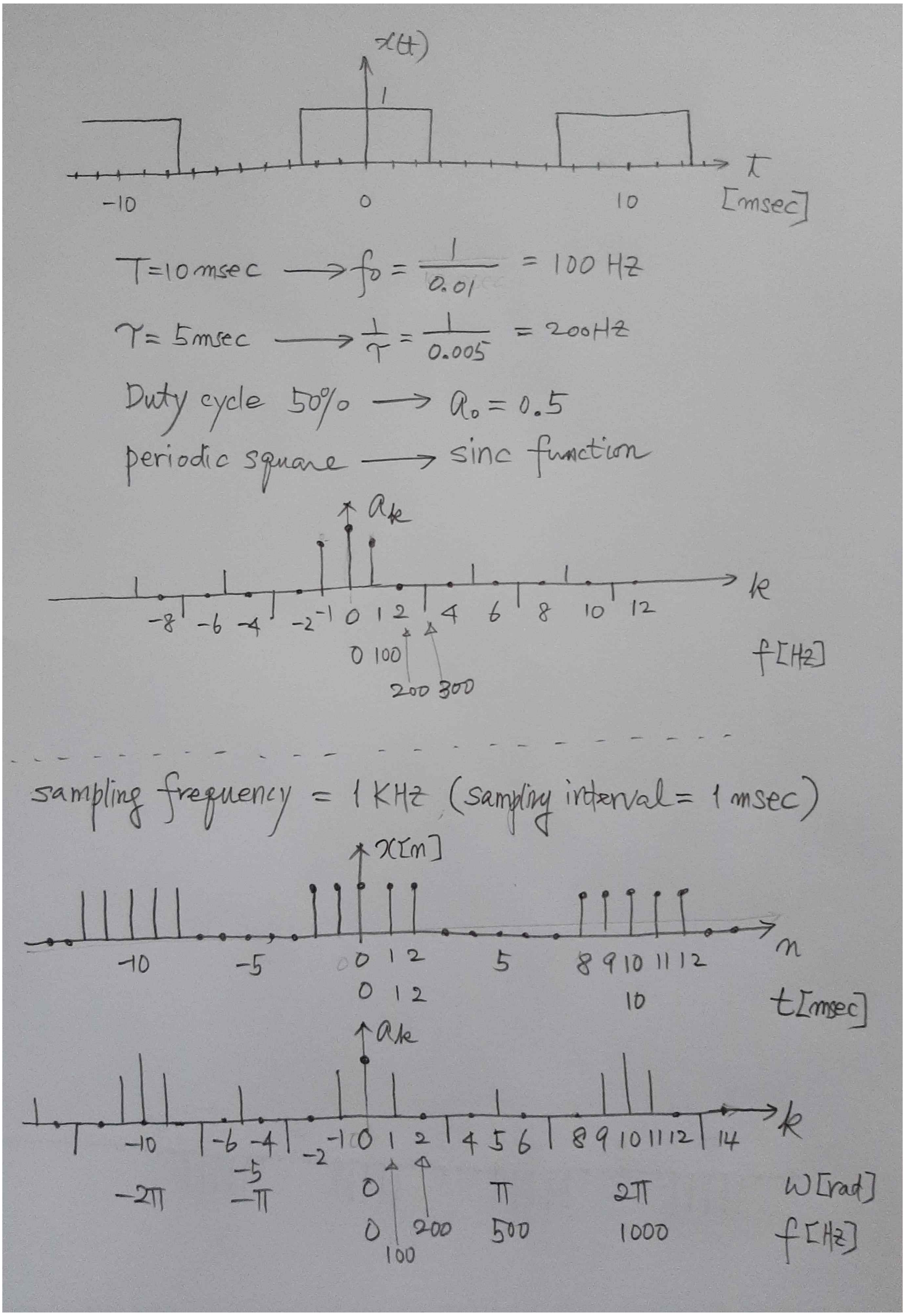
$$= \frac{1}{N} \frac{e^{-jk(2\pi/2N)}\left[e^{jk2\pi(N_1+1/2)/N} - e^{-jk2\pi(N_1+1/2)/N}\right]}{e^{-jk(2\pi/2N)}\left[e^{jk(2\pi/2N)} - e^{-jk(2\pi/2N)}\right]}$$

$$= \frac{1}{N} \frac{\sin[2\pi k(N_1+1/2)/N]}{\sin(\pi k/N)}, \quad k \neq 0, \pm N, \pm 2N, \cdots \quad (3.104)$$

and $a_k = \dfrac{2N_1+1}{N}, \quad k = 0, \pm N, \pm 2N, \cdots$ (3.105)

a_k for $2N_1+1=5$ and $N=10$	
a_k for $2N_1+1=5$ and $N=20$	
a_k for $2N_1+1=5$ and $N=40$	

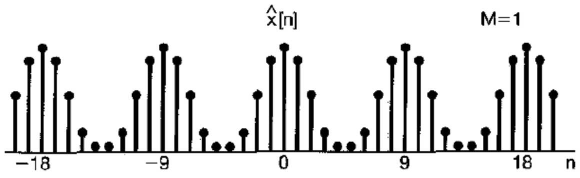
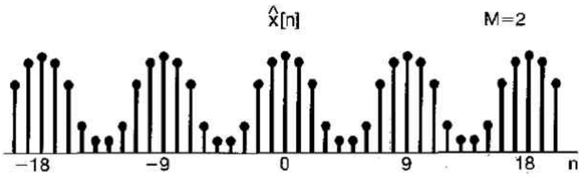
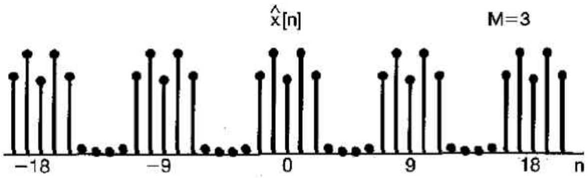
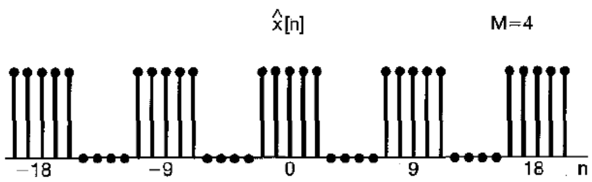
- Comparison CTFS / DTFS (sampling)



Synthesis by using the partial sum of the harmonics

$$\hat{x}[n] = \sum_{k=-M}^{M} a_k e^{jk(2\pi/N)n}$$

(Example) $N=9,\ 2N_1+1=5$ and for several values of M

$M=1$	
$M=2$	
$M=3$	
$M=4$	

※ For $M=4$, the partial sum $\hat{x}[n]$ exactly equals $x[n]$.

- In contrast to the continuous-time case, there are NO convergence issues and there is NO Gibbs phenomenon, since $x[n]$ is completely specified by a finite number N of parameters.
- Since a continuous-time periodic signal takes on a continuum of values over a single period, and an infinite number of Fourier coefficients are required to represent the signal.

(Example) Electrocardiogram(ECG) waveform

- Normal ECG / Ventricular Tachycardia ECG
  (time-domain waveforms and their spectra)

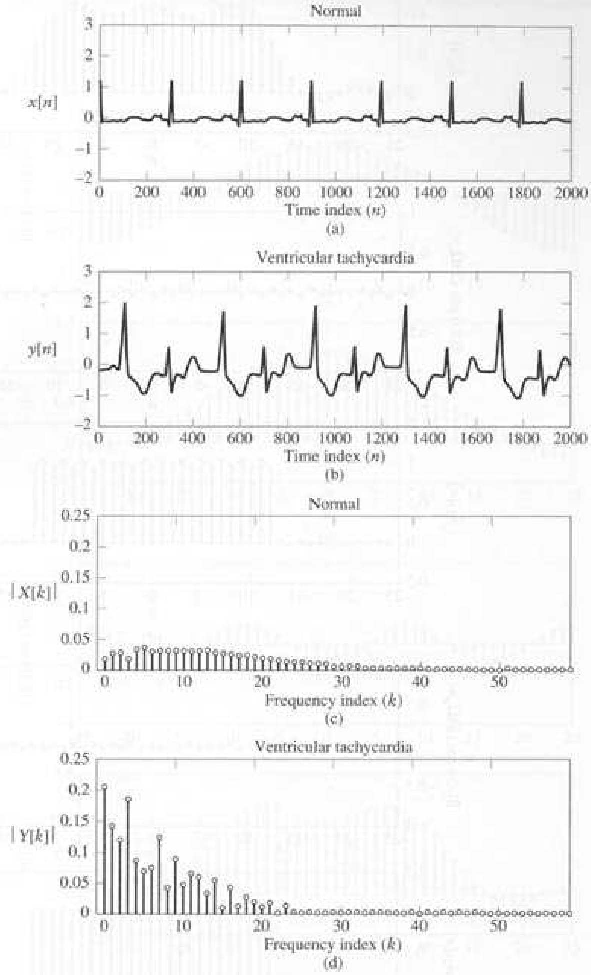


FIGURE 3.7 Electrocardiograms for two different heartbeats and the first 60 coefficients of their magnitude spectra. (a) Normal heartbeat. (b) Ventricular tachycardia. (c) Magnitude spectrum for the normal heartbeat. (d) Magnitude spectrum for ventricular tachycardia.

※ Spectral Analysis

We can get more information
from the spectrum in the frequency domain
than from the waveform in the time domain.

3.7 Properties of DTFS

TABLE 3.2 PROPERTIES OF DISCRETE-TIME FOURIER SERIES

Property	Periodic Signal	Fourier Series Coefficients
	$\left.\begin{array}{l} x[n] \\ y[n] \end{array}\right\}$ Periodic with period N and fundamental frequency $\omega_0 = 2\pi/N$	$\left.\begin{array}{l} a_k \\ b_k \end{array}\right\}$ Periodic with period N
Linearity	$Ax[n] + By[n]$	$Aa_k + Bb_k$
Time Shifting	$x[n - n_0]$	$a_k e^{-jk(2\pi/N)n_0}$
Frequency Shifting	$e^{jM(2\pi/N)n} x[n]$	a_{k-M}
Conjugation	$x^*[n]$	a^*_{-k}
Time Reversal	$x[-n]$	a_{-k}
Time Scaling	$x_{(m)}[n] = \begin{cases} x[n/m], & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$ (periodic with period mN)	$\frac{1}{m} a_k \begin{pmatrix}\text{viewed as periodic} \\ \text{with period } mN\end{pmatrix}$
Periodic Convolution	$\sum_{r=\langle N\rangle} x[r]y[n-r]$	$Na_k b_k$
Multiplication	$x[n]y[n]$	$\sum_{l=\langle N\rangle} a_l b_{k-l}$
First Difference	$x[n] - x[n-1]$	$(1 - e^{-jk(2\pi/N)})a_k$
Running Sum	$\sum_{k=-\infty}^{n} x[k] \begin{pmatrix}\text{finite valued and periodic only} \\ \text{if } a_0 = 0\end{pmatrix}$	$\left(\frac{1}{(1 - e^{-jk(2\pi/N)})}\right)a_k$
Conjugate Symmetry for Real Signals	$x[n]$ real	$\begin{cases} a_k = a^*_{-k} \\ \mathcal{R}e\{a_k\} = \mathcal{R}e\{a_{-k}\} \\ \mathcal{I}m\{a_k\} = -\mathcal{I}m\{a_{-k}\} \\ \lvert a_k\rvert = \lvert a_{-k}\rvert \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	$x[n]$ real and even	a_k real and even
Real and Odd Signals	$x[n]$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\begin{cases} x_e[n] = \mathcal{E}v\{x[n]\} & [x[n] \text{ real}] \\ x_o[n] = \mathcal{O}d\{x[n]\} & [x[n] \text{ real}] \end{cases}$	$\mathcal{R}e\{a_k\}$ $j\mathcal{I}m\{a_k\}$

Parseval's Relation for Periodic Signals

$$\frac{1}{N}\sum_{n=\langle N\rangle} |x[n]|^2 = \sum_{k=\langle N\rangle} |a_k|^2$$

※ Notation of DTFS

$$x[n] \quad \leftrightarrow \quad a_k$$
$$y[n] \quad \leftrightarrow \quad b_k$$

§3.7.1 Multiplication

$$x[n]\,y[n] \qquad \leftrightarrow \qquad d_k = \sum_{l=<N>} a_l\, b_{k-l} \tag{3.108}$$

$$\begin{pmatrix}\text{multiplication} \\ \text{in time domain}\end{pmatrix} \qquad \begin{pmatrix}\text{convolution} \\ \text{in frequency domain}\end{pmatrix}$$

§3.7.2 First Difference

$$x[n]-x[n-1] \quad \leftrightarrow \quad (1-e^{-jk(2\pi/N)})\ a_k$$

※ (time shifting in time domain)≡ (phase shifting in freq. domain)

(Example 3.13) Find a_k for $x[n]$

$$x[n]=x_1[n]+x_2[n]$$

$$x_1[n] \quad \leftrightarrow \quad b_k$$

$$x_2[n] \quad \leftrightarrow \quad c_k$$

$$a_k=b_k+c_k$$

§3.8 Fourier Series and LTI Systems

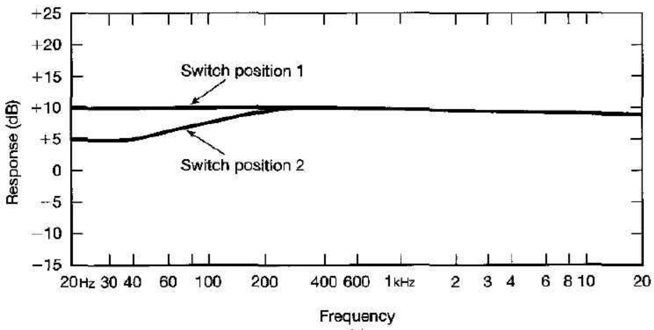
		continuous LTI system	discrete LTI system
single complex exponential	input	$x(t)=e^{st}$	$x[n]=z^n$
	impulse response	$h(t)$	$h[n]$
	output	$y(t)=H(s)\,e^{st}$	$y[n]=H(z)\,z^n$
	transfer function	$H(s)=\int_{-\infty}^{+\infty} h(t)\,e^{-st}\,dt$	$H(z)=\sum_{n=-\infty}^{+\infty} h[n]\,z^{-n}$
	freq. response	$H(j\omega)=\int_{-\infty}^{+\infty} h(t)\,e^{-j\omega t}\,dt$	$H(e^{j\omega})=\sum_{n=-\infty}^{+\infty} h[n]\,e^{-j\omega n}$
linear combination of complex exponentials	FS for a periodic signal	$x(t)=\sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$	$x[n]=\sum_{k=<N>} a_k e^{jk\omega_0 n}$
	output	$y(t)=\sum_{k=-\infty}^{+\infty} a_k\,H(jk\omega_0)\,e^{jk\omega_0 t}$	$y[n]=\sum_{k=<N>} a_k H(e^{jk\omega_0})\,e^{jk\omega_0 n}$
	remark	The effect of the LTI system is to modify individually each of the Fourier coefficients of the input through multiplication by the value of the frequency response of the LTI system, $H(jk\omega_0)$, at the corresponding frequency.	The effect of the LTI system is to modify individually each of the Fourier coefficients of the input through multiplication by the value of the frequency response of the LTI system, $H(e^{j2\pi k/N})$, at the corresponding frequency.
	read	Nilsson, Electric Circuits §14.2	
	example	Example 3.16 (≡ Nilsson Ex14.2)	Example 3.17

3.9 Filtering

3.9.1 Frequency-Shaping Filter

(example 1) audio systems

- modify the relative amounts of low-frequency energy (bass) and high-frequency energy (treble)

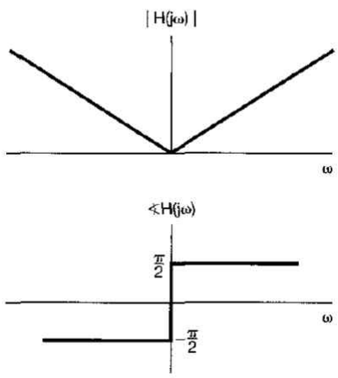


(example 2) a differentiating filter (edge detector): $y(t) = \dfrac{dx(t)}{dt}$

[difference in the discrete-time domain]

- a high-pass filter
- If $x(t) = e^{j\omega t}$, then the output is $y(t) = j\omega\, e^{j\omega t}$.
- the frequency response

$$H(j\omega) = j\omega$$



- Effects : Fig. 3.24 (edge detection)

※ blurring : low-pass filter (≡ integration)

[accumulation (i.e., addition) in the discrete-time domain]

(example) $H(j\omega) = \dfrac{1}{j\omega}$

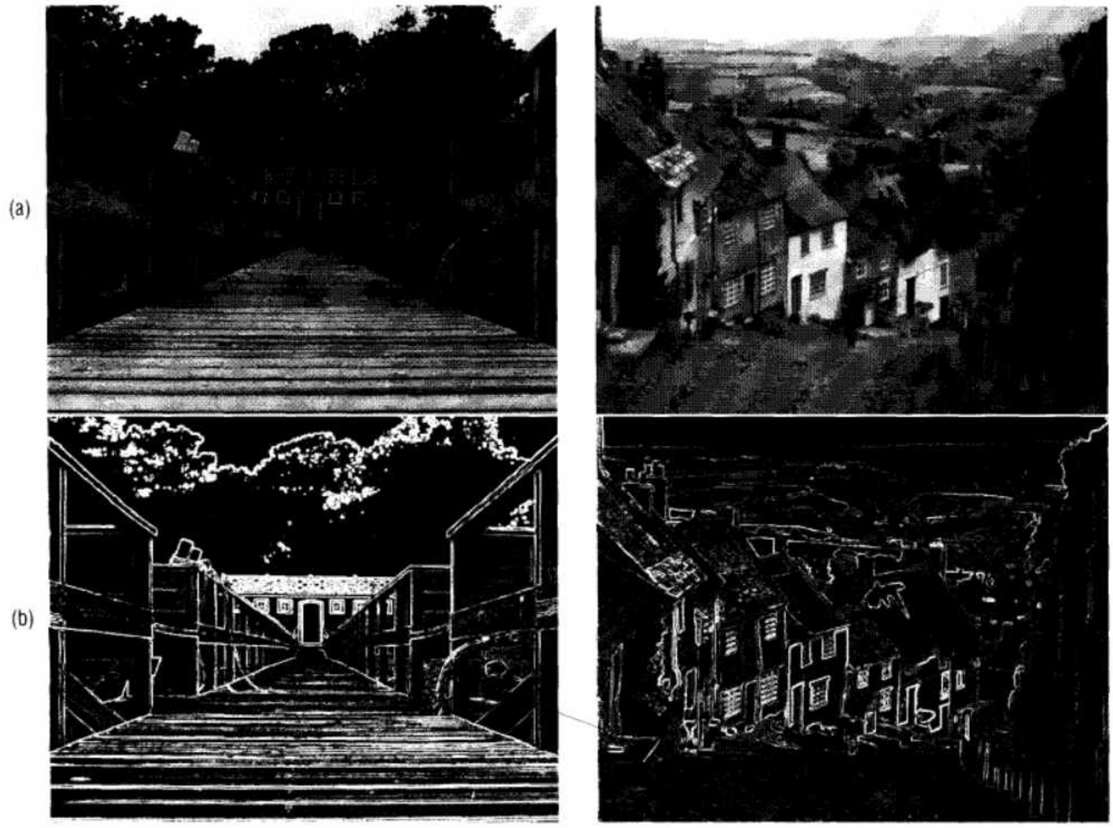
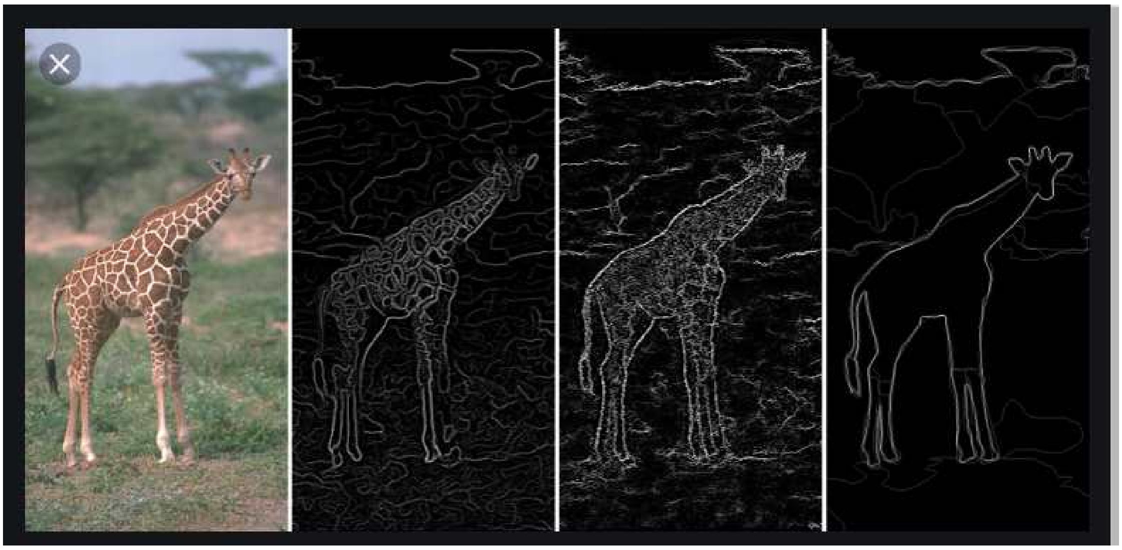
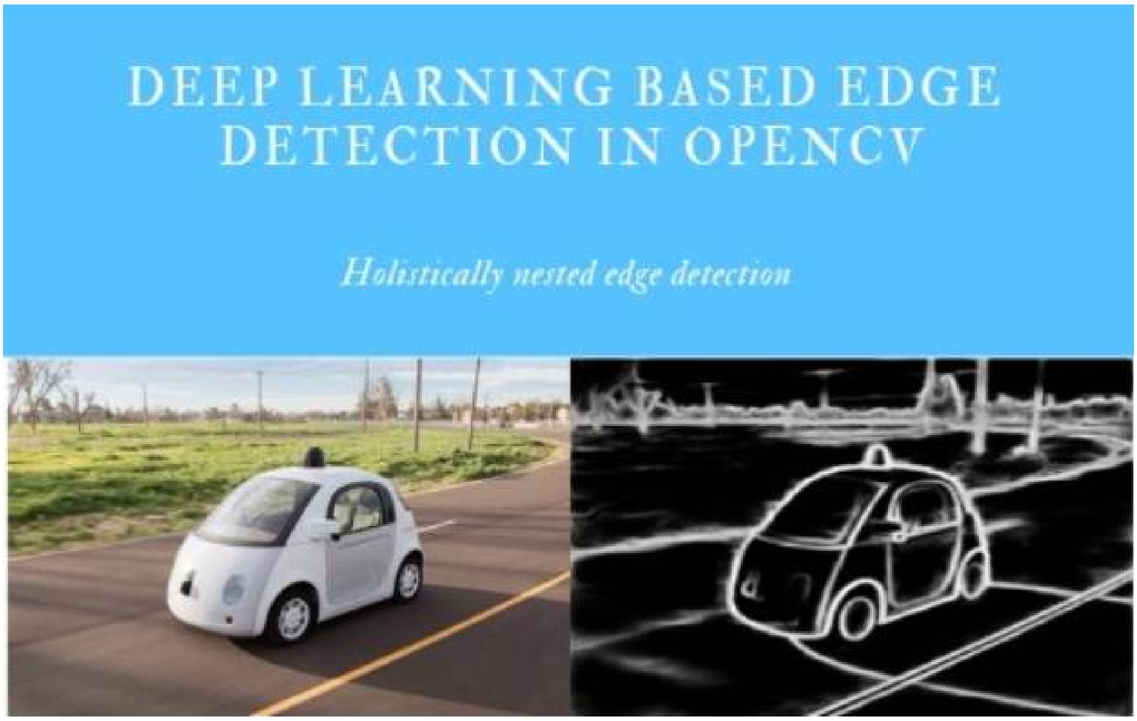


Figure 3.24 Effect of a differentiating filter on an image: (a) two original images; (b) the result of processing the original images with a differentiating filter.



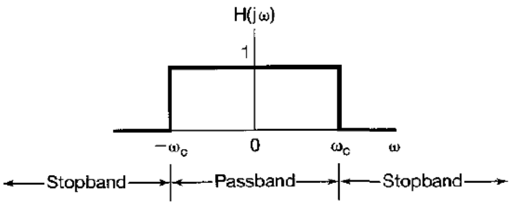


※ OpenCV (Open source computer vision)

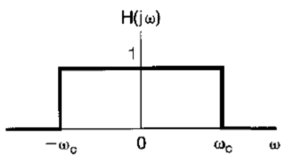
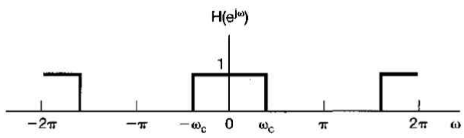
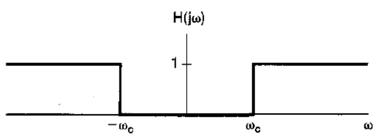
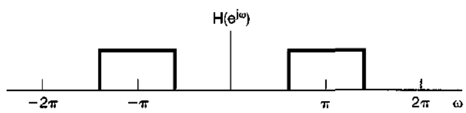
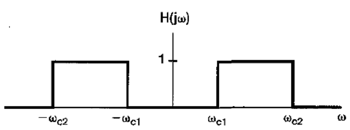
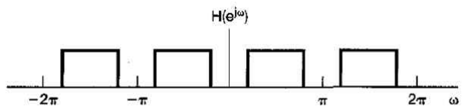
3.9.2 Frequency-Selective Filters (Nilsson, Chapter 14)

- low-pass filter (LPF), high-pass filter (HPF), bandpass filter(BPF)
- cutoff frequency, passband, stopband
- continuous-time ideal low-pass filter

$$H(j\omega) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & |\omega| > \omega_c \end{cases} \qquad (3.140)$$



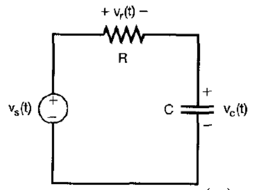
Ideal Filters

	continuous-time	discrete-time
ideal LPF		
ideal HPF		
ideal BPF		

3.10 Examples of CT filters described by Differential Equations

3.10.1 A simple RC LPF (Nilsson, Ex.14.2, exactly same example)

※ Nilsson §7.3 (step functions), §9.2 (sinusoids), §14.2



$$RC\,\frac{dv_c(t)}{dt} + v_c(t) = v_s(t) \tag{3.141}$$

If the input is $v_s(t) = e^{j\omega t}$,

then we must have the output voltage $v_c(t) = H(j\omega)\, e^{j\omega t}$

$$RC\,\frac{d}{dt}[H(j\omega)\, e^{j\omega t}] + H(j\omega)\, e^{j\omega t} = e^{j\omega t} \tag{3.142}$$

$$RCj\omega\, H(j\omega)\, e^{j\omega t} + H(j\omega)\, e^{j\omega t} = e^{j\omega t} \tag{3.143}$$

$$H(j\omega)\, e^{j\omega t} = \frac{1}{1+RCj\omega}\; e^{j\omega t} \tag{3.144}$$

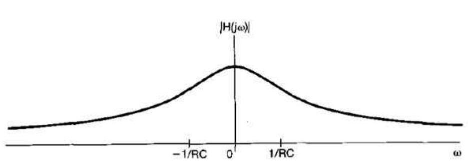
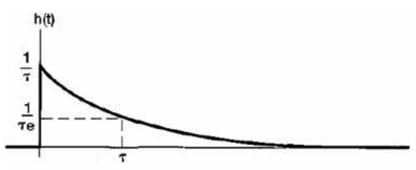
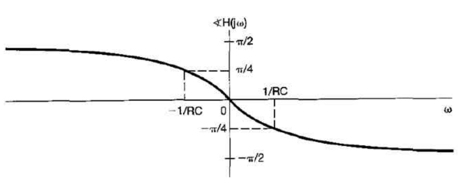
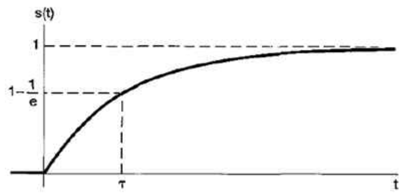
Therefore the frequency response is

$$H(j\omega) = \frac{1}{1+RCj\omega} \tag{3.145}$$

Solving Eq.(3.141), we can obtain the impulse response $h(t)$,

$$h(t) = \frac{1}{RC}\, e^{-t/RC}\, u(t) \tag{3.146}$$

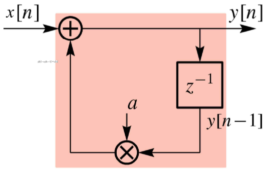
※ Nilsson Eq.(14.7-10), Ex.14.2, §13.8, Eq.(7.25), Eq.(7.51)

amplitude and phase of the frequency response (Nilsson Fig.14.5)	impulse response / step response (Nilsson Fig.13.49 / Fig.7.17)
	
	

3.11 Examples of DT filters described by Difference Equations

3.11.1 First-Order Recursive Discrete-Time filters

(Example) $y[n] - ay[n-1] = x[n]$ (3.151)



$$y[n] = x[n] + ay[n-1]$$

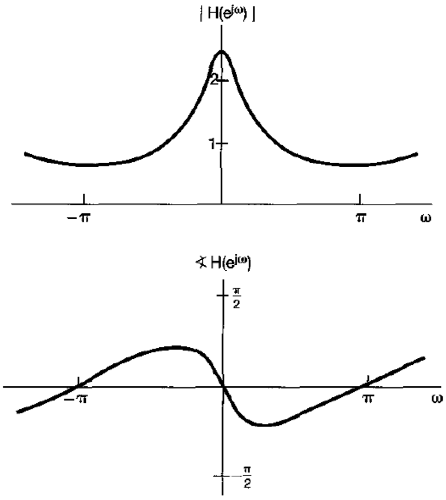
If the input is $x[n] = e^{j\omega n}$,
then we must have the output voltage $y[n] = H(e^{j\omega})\, e^{j\omega n}$,
where $H(e^{j\omega})$ is the frequency response of the system.

$$H(e^{j\omega})\, e^{j\omega n} - aH(e^{j\omega})\, e^{j\omega (n-1)} = e^{j\omega n} \qquad (3.152)$$

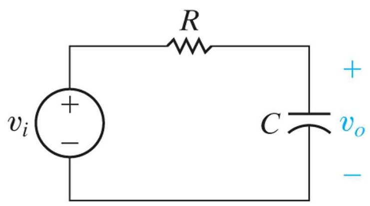
$$\therefore \quad H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}} \qquad (3.154)$$

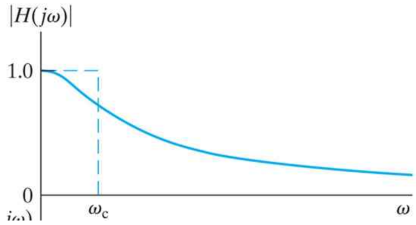
※ a digital Infinite Impulse Response (IIR) LPF

(Example) $a = 0.6$



※ analog passive LPF





impulse response : $h[n] = a^n u[n]$

step response : $s[n] = \dfrac{1 - a^{n+1}}{1-a}\, u[n]$

※ See Eq.(3.145), (3.146) and (3.147).

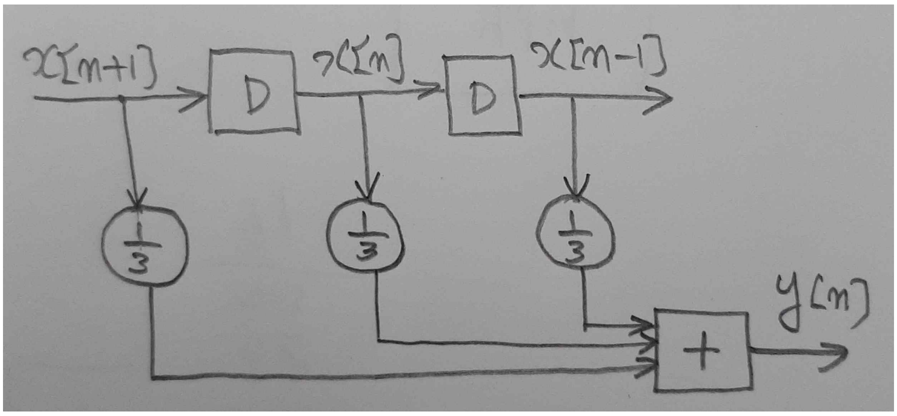
3.11.2 Nonrecursive Discrete-Time filters

$$y[n] = \sum_{k=-N}^{M} b_k\, x[n-k] \tag{3.157}$$

※ moving average (MA) filter
running average filter

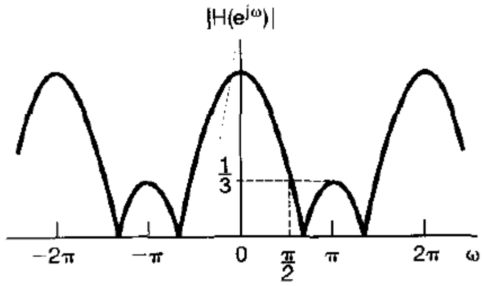
※ weighted sum of inputs

(Example) $y[n] = \frac{1}{3}\,(x[n-1]+x[n]+x[n+1])$

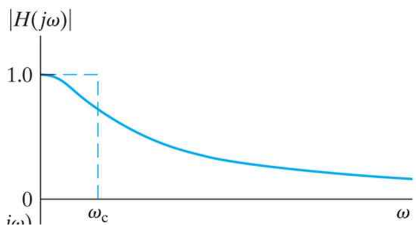


$$h[n] = \frac{1}{3}\,(\delta[n-1]+\delta[n]+\delta[n+1])$$

$$H(e^{j\omega}) = \frac{1}{3}\,(e^{j\omega}+1+e^{-j\omega}) = \frac{1}{3}(1+2\cos\omega)$$



※ a digital nonrecursive LPF
(a digital Finite Impulse Response
(FIR) LPF)

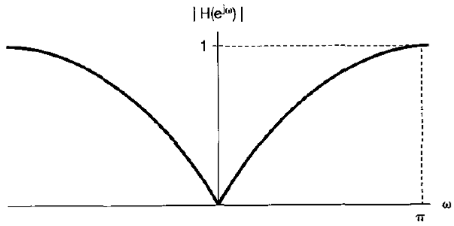


※ an analog LPF

(Example) Difference [High Pass Filter]

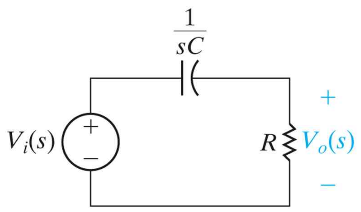
$$y[n] = \frac{x[n] - x[n-1]}{2} \tag{3.163}$$

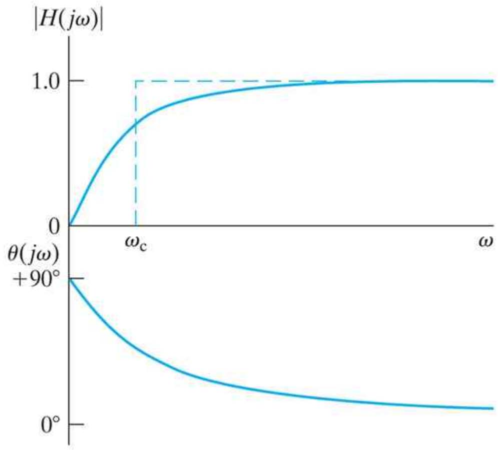
$$H(e^{j\omega}) = \frac{1}{2}\left(1 - e^{-j\omega}\right) = j\,e^{j\omega/2}\sin(\omega/2) \tag{3.164}$$



※ edge detector

(※ Circuit and Lab Chapter 14)

	$H(j\omega) = \frac{V_0(j\omega)}{V_i(j\omega)} = \frac{j\omega}{j\omega + 1/RC}$



3.12 Summary

- Fourier representation

	periodic function	nonperiodic function
continuous time function	continuous-time Fourier Series ((CT)FS)	continuous-time Fourier transform ((CT)FT)
discrete time function	discrete-time Fourier Series (DTFS)	discrete-time Fourier transform (DTFT)

- representation methods of square waves

Fig. 3.6 : waveform of a periodic continuous-time square wave
Fig. 3.7 : spectrum of a periodic continuous-time square wave
Fig. 3.16 : waveform of a periodic discrete-time square wave
Fig. 3.17 : spectrum of a periodic discrete-time square wave

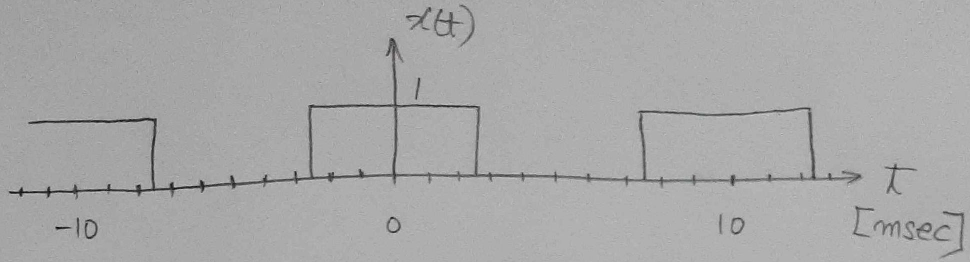
(Example 3.5)
Fourier series of a periodic continuous-time square wave
(Example 3.12)
Fourier series of a periodic discrete-time square wave

※ Compare Examples 3.5 and 3.12 (Figs. 3.6, 3.7, 3.16, 3.17)

sifting property (chap. 2)	Fourier series (Chap. 3)
a linear combination of weighted impulses in the time domain	a linear combination of weighted impulses in the frequency domain
$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$ ※ $\sum_{k=-\infty}^{\infty} a_k \delta[\omega - k\omega_0]$

- Linear Combination representation for LTI system

※ A sine wave can be represented by using impulse in the frequency domain.

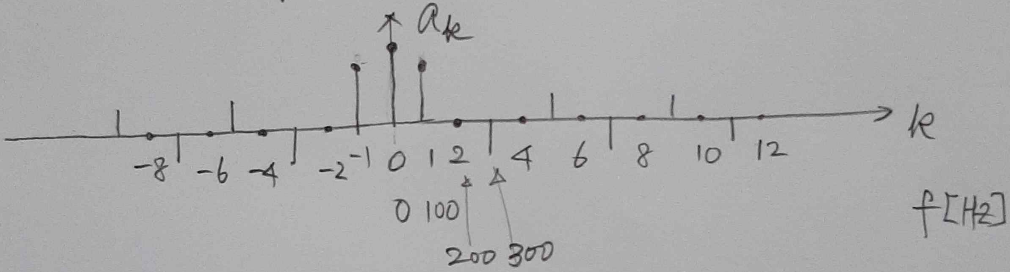


$T = 10\,\text{msec} \longrightarrow f_0 = \frac{1}{0.01} = 100\,\text{Hz}$

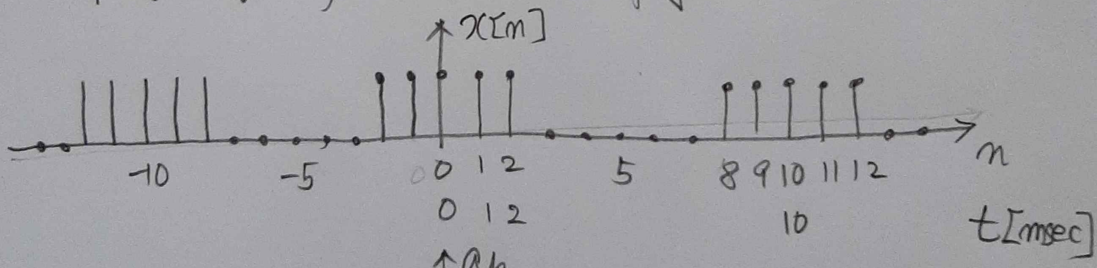
$\tau = 5\,\text{msec} \longrightarrow \frac{1}{\tau} = \frac{1}{0.005} = 200\,\text{Hz}$

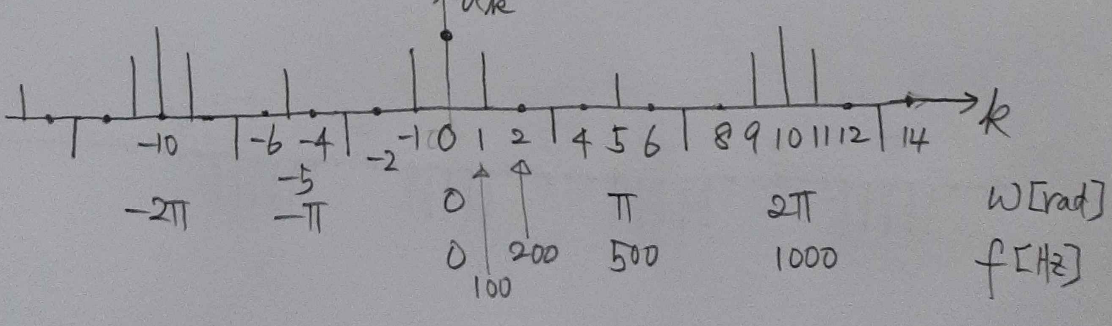
Duty cycle 50% $\longrightarrow a_0 = 0.5$

periodic square $\longrightarrow$ sinc function



sampling frequency = 1 KHz (sampling interval = 1 msec)





(Example) RC circuit with a periodic square wave

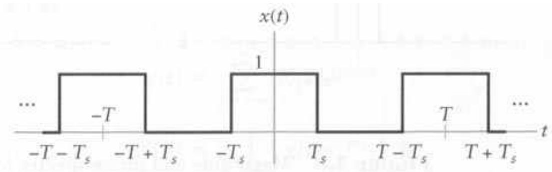
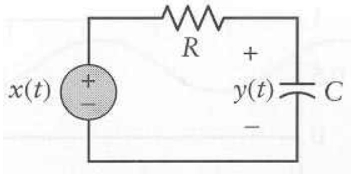
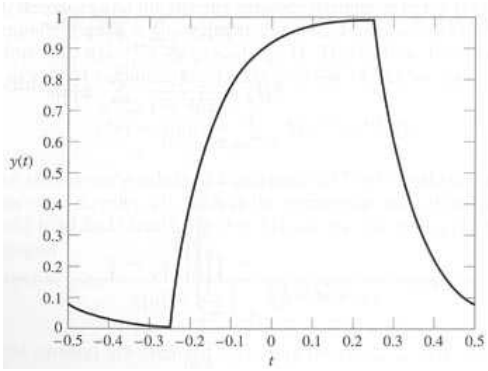


FIGURE 3.9 Square wave for Example 3.6.



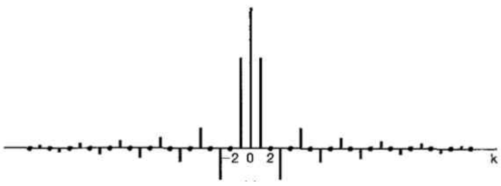
$$T = 1\,\text{sec}, \quad \frac{T_s}{T} = \frac{1}{4}, \quad RC = 0.1\,\text{sec}$$

- output signal in the time domain



- input spectrum and output spectrum

(spectrum of input)



(spectrum of output)